Product Recommendation Chatbot

Design Choices

- Parent and child flows: Main code calls on a separate workflow depending on user input.
- Retry counter: Implemented a counter to prevent infinite loops with red text to show visual urgency.
- Multiple inputs accepted: The user can enter either a word (e.g. macOS) or a number (e.g. 2) to achieve the same result

Technical Implementation

- CLI interface with Python3
- No external libraries needed
- Numbers and words are associated with each other through
- .strip() and .lower() are used to forcibly change user inputs in the back end to resolve case sensitivity issues

Challenges faced

- Limitations on the amount of options the chatbot could recommend
- Matching the word and number inputs required duplicating every valid option in the list (e.g. ['windows', 'macos', '1', '2']). This works but is verbose — a cleaner abstraction would reduce repetition.
- Implementing .lower() to user input caused bugs where the child flow did not occur correctly, necessitating consistent lowercase spelling throughout all of the code (see right)

```
Hello! Welcome to the chatbot. Let me help you find a device.

What type of product are you looking for today?
 1) laptop
  2) phone
Your answer: 2
Understood. Let me ask you a few questions first.
Do you prefer Android or iPhone?
  1) Android
  2) iPhone
Your answer: iphone
That is not a valid option. Please choose from one of the following options: ['Android
', 'iPhone', '1', '2']

You have 3 attempts left.
Do you prefer Android or iPhone?
  1) Android
  2) iPhone
Your answer: iPhone
That is not a valid option. Please choose from one of the following options: ['Android
', 'iPhone', '1', '2']

You have 2 attempts left.
Do you prefer Android or iPhone?
  1) Android
  2) iPhone
Your answer: Iphone
That is not a valid option. Please choose from one of the following options: ['Android
', 'iPhone', '1', '2']

You have 1 attempts left.
Do you prefer Android or iPhone?
  1) Android
  2) iPhone
Your answer: 2
What is your budget?
  1) low (under $500)
```

Suggested improvements

- Add API to fetch the most up-to-date monetary values and availability of certain devices
- Add more options for branching paths (e.g. tablets, phone plans, phones that are not Android or iOS)
- Have the program ask if the user would like to search for another product at the end of each session
- Allow user to add their own budget number instead of relying on rough estimates (low vs. high)
- Experiment more with colored text so that chatbot text and user input are more easily distinguishable